AYE

• ELECTRICAL ENGINEERING • LESSON 09

Circuit Design
Basics

Ohm's Law | Series/Parallel | KVL | KCL | Power | Protection

What You Will Learn

- Ohm's Law and the three power formulas — applied to real circuit design decisions
- Series and parallel circuit analysis — resistance, voltage, current, power in each
- Kirchhoff's Voltage Law (KVL) and Current Law (KCL) — the tools for any circuit
- Voltage dividers, protection design, and fuse/MCB selection for practical circuits

Ohm's Law Triangle

$V = I \times R$ | $I = V/R$ | $R = V/I$

V = Voltage (Volts) I = Current (Amperes) R = Resistance (Ohms)

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ELEC-009: Circuit Design Basics

Understanding how electrical circuits work — how voltage, current, and resistance relate to each other, and how power is dissipated — is the foundation of every electrical engineering decision. Every cable sizing calculation, every fuse selection, every motor starter design begins with these fundamentals. This lesson establishes the core analytical tools that underpin all practical electrical engineering work.

Learning Objectives

- Apply Ohm's Law to calculate voltage, current, or resistance in any circuit element
- Calculate power dissipation using all three power formulas and identify the correct formula for given information
- Analyse series and parallel circuits: total resistance, branch currents, voltage drops
- Apply KVL and KCL to solve circuits with multiple loops and junction nodes
- Design a voltage divider and select protection devices (fuses, MCBs) for a given circuit

1. Ohm's Law — The Foundation of All Circuit Analysis

Ohm's Law states that the voltage across a resistive element equals the current through it multiplied by its resistance. This simple relationship governs almost every practical electrical calculation.

$$V = I \times R \mid I = V / R \mid R = V / I$$

Quantity	Symbol	SI Unit	Practical Definition
Voltage (V)	V or U	Volt (V)	The electrical "pressure" that drives current through a circuit. A 230V supply has 230V potential difference between line and neutral.
Current (I)	I	Ampere (A)	The rate of flow of electric charge. A 13A socket can supply up to 13 coulombs of charge per second.
Resistance (R)	R	Ohm (Ω)	Opposition to current flow. A 1000W electric heater at 230V: $R = V^2/P = 230^2/1000 = 52.9\Omega$
Power (P)	P	Watt (W)	Rate of energy conversion. A 100W lamp converts 100 joules of electrical energy to light and heat every second.

2. Power Formulas — Three Ways to Calculate

$$P = V \times I$$

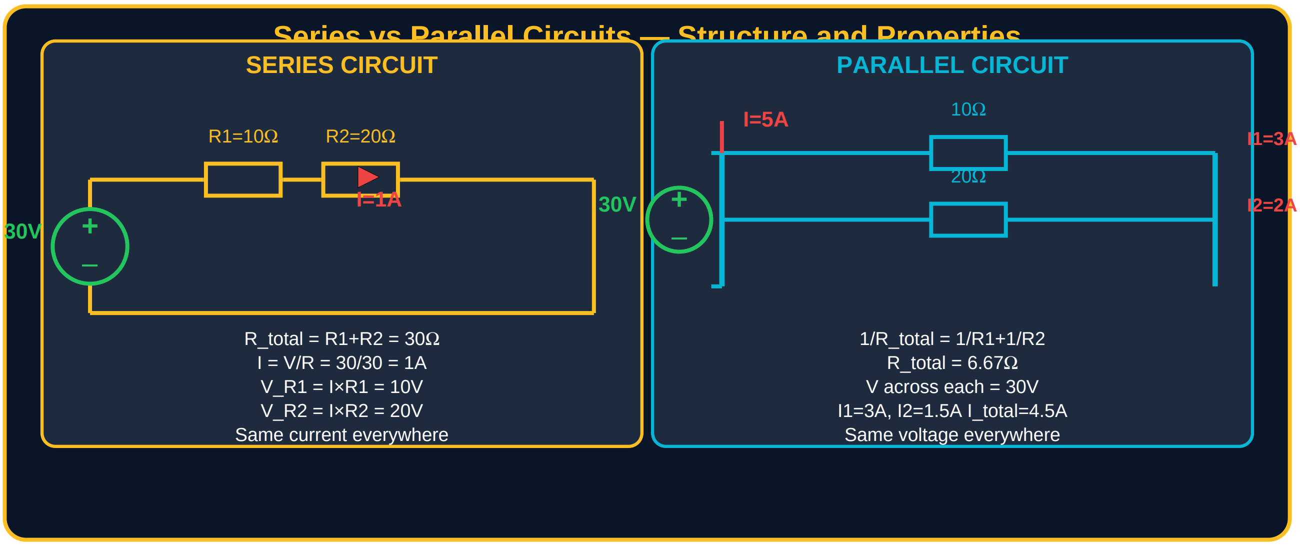
$$P = I^2 \times R$$

$$P = V^2 / R$$

Choose the formula based on what information is available. All three give the same result if the values are consistent.

Given	Formula	Example	Answer
V and I known	$P = V \times I$	230V supply, 4.35A current	$P = 230 \times 4.35 = 1{,}000W = 1\ kW$
I and R known	$P = I^2 \times R$	5A through a 20Ω resistor	$P = 5^2 \times 20 = 25 \times 20 = 500W$
V and R known	$P = V^2 / R$	12V across a 6Ω resistance	$P = 12^2 / 6 = 144 / 6 = 24W$
P and V known	$I = P / V$	2,300W load at 230V	$I = 2{,}300 / 230 = 10A$
P and I known	$R = P / I^2$ or $V = P/I$	500W, 2A	$R = 500 / 4 = 125\Omega$; $V = 500/2 = 250V$

3. Series and Parallel Circuits

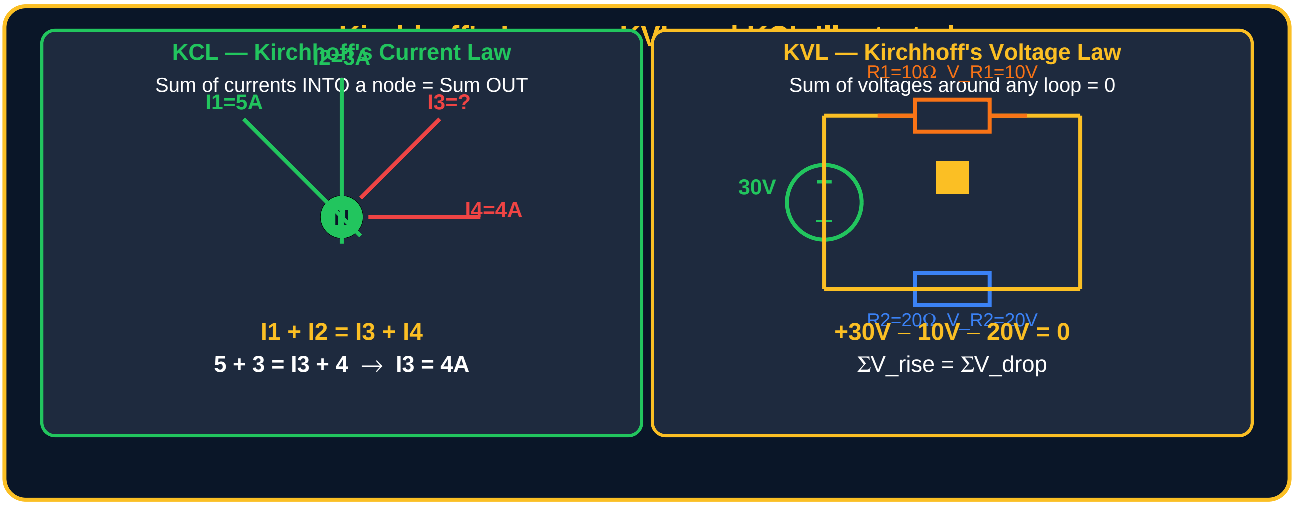


Series vs Parallel — Design Implications

Property	Series Circuit	Parallel Circuit	Design Use
Total resistance	R_total = R1 + R2 + … (always greater than any individual R)	1/R_total = 1/R1 + 1/R2 + … (always less than smallest R)	Series: increase resistance. Parallel: decrease resistance.
Current	Same through all elements — I is equal everywhere	Splits between branches. Each branch carries independently determined current.	Parallel feeds: each load gets independent current. Series: overload one = overload all.
Voltage	Divides across elements. V_R = I × R for each.	Same across all parallel branches — equal to supply voltage.	Parallel loads: all at supply voltage. Series: voltage divides (used intentionally in voltage dividers).
Failure effect	One open circuit = entire circuit fails (Christmas lights old-style)	One open branch = other branches continue normally	Building circuits: parallel for reliability. Series: for control (E-stop in series stops all).
Power	P_total = P1 + P2 + … Power in each = $I^2 \times R_n$	P_total = P1 + P2 + … Power in each = V^2 / R_n	Total power always sums regardless of connection type.

4. Kirchhoff's Laws — Solving Any Circuit

Ohm's Law applies to single elements. Kirchhoff's Laws allow you to analyse complete circuits with multiple sources, multiple loops, and multiple junction nodes. Together, KVL and KCL can solve any linear resistive network.



How to Apply KVL — Step by Step

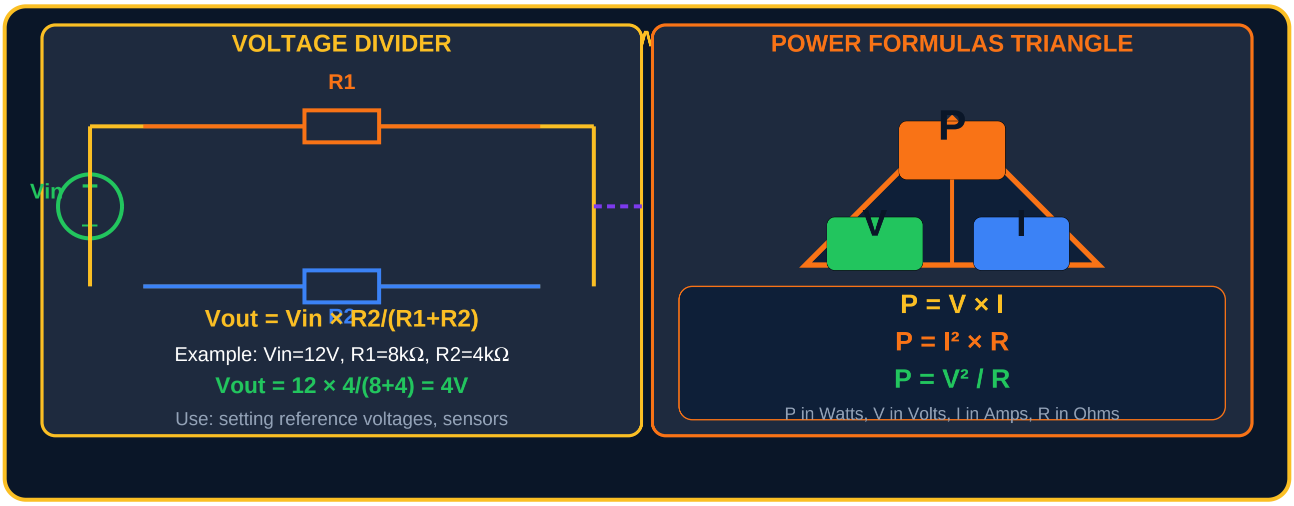
Step	Action	Example (single loop, 30V source, R1=10Ω, R2=20Ω)
1	Assign current direction (arbitrary — negative result means opposite direction)	Assume current flows clockwise, call it I
2	Traverse the loop, noting voltage rises (+) and drops (–)	Going clockwise: +30V (source), –V_R1, –V_R2
3	Apply KVL: sum of all voltage terms = 0	+30 – I×10 – I×20 = 0
4	Solve for unknown	30 – 30I = 0 → I = 1A
5	Calculate individual voltages	V_R1 = 1×10 = 10V; V_R2 = 1×20 = 20V. Check: 10+20 = 30V ✓

KCL — Node Voltage Method

KCL (currents into a node = currents out) is applied at every node except the reference (ground) node. Writing KCL equations for all independent nodes gives a system of equations that can be solved simultaneously for all unknown node voltages. This is the most systematic method for complex multi-loop circuits.

- **Step 1:** Label all nodes. Choose one as reference (ground, V=0).
- **Step 2:** At each non-reference node, write KCL: sum of currents leaving = 0.
- **Step 3:** Express each current in terms of node voltages using Ohm's Law: $I = (V_{n1} - V_{n2})/R$.
- **Step 4:** Solve the resulting system of equations simultaneously.
- **Step 5:** Calculate all branch currents and power from the node voltages.

5. Voltage Divider and Power Formulas



6. Circuit Protection Design — Fuse and MCB Selection

Every electrical circuit requires protection against overload and short circuit. The protective device (fuse or MCB) must be rated to:
(1) carry the normal operating current without tripping, (2) trip quickly on fault current, and (3) not trip on harmless transients (motor start surges, transformer inrush).

$$I_{fuse} \text{ or } I_{MCB} \geq I_{design} \text{ AND } \leq I_{max\_cable}$$

Protection Device	Rated Current Range	Breaking Capacity	Trip Characteristics	Application
Glass fuse (cartridge)	0.1A to 100A	1,500A (domestic)	Operates on I^2t — slower at low overloads, fast on short circuit	Domestic equipment, electronics. One-shot — replace after operation.
MCB Type B	6A to 63A	6,000A (BS EN 60898)	Trips at 3–5× rated current. Fast for modest overloads.	Resistive loads: lighting, heating, sockets. Most common domestic MCB.
MCB Type C	6A to 63A	6,000A (BS EN 60898)	Trips at 5–10× rated current. Tolerates higher inrush.	Inductive loads: motors, transformers, fluorescent lighting with magnetic ballasts.
MCB Type D	6A to 125A	6,000A (BS EN 60898)	Trips at 10–20× rated current. Maximum transient tolerance.	High inrush: welding equipment, X-ray machines, large transformers.

MCCB	16A to 1,600A	25,000–100,000A	Adjustable trip settings. Thermal-magnetic or electronic trip units.	Industrial, commercial, distribution boards. Larger circuits than MCBs.

Fuse and MCB Selection — Worked Example

Load: 2.2 kW single-phase motor at 230V, power factor 0.85. Start-up current = 6× FLA. Cable rated at 20A.
FLA = P / (V × PF) = 2,200 / (230 × 0.85) = **11.3A**
Start-up current = 6 × 11.3 = **67.8A** (momentary)
Select MCB: must carry 11.3A continuously → choose **16A Type C MCB**. At start-up: 16A × 5 (min Type C trip) = 80A. 67.8A < 80A → does NOT trip on start ✓. Cable (20A) > MCB (16A) → cable protected ✓

7. Practice Exercises

#	Exercise	Data / Instructions
1	Ohm's Law applications	A 230V circuit has a 46Ω heater element. (a) Calculate the current drawn. (b) Calculate the power consumed (use all three power formulas — verify all give the same result). (c) What resistance would draw exactly 10A at 230V? What power would this dissipate?
2	Series circuit analysis	Three resistors in series: R1=15Ω, R2=25Ω, R3=60Ω, connected to 100V supply. (a) Total resistance. (b) Current. (c) Voltage across each resistor. (d) Power dissipated in each resistor. (e) Total power from supply. (f) Verify using P_total = V_supply × I.
3	Parallel circuit analysis	Three parallel loads on a 230V supply: Load1=1150W, Load2=2300W, Load3=575W. (a) Current drawn by each load. (b) Resistance of each load. (c) Total current from supply. (d) Total equivalent resistance. (e) Total power. (f) Which load draws most current?
4	KVL application	A circuit has a 48V supply, R1=12Ω and R2=4Ω in series, with a 24V battery (opposing direction) in series with R2. Apply KVL to find the current in the circuit. Then calculate voltage across R1 and R2. Verify KVL holds around the loop.
5	Protection device selection	A domestic ring final circuit supplies: 3 sockets in use simultaneously at 700W each, plus a 3kW kettle. Supply: 230V single phase. (a) Calculate total load current. (b) Select the correct MCB rating and type. (c) The cable is $2.5mm^2$ rated 23A. Is the cable adequately protected?

Lesson Summary

Concept	Key Formula / Rule
Ohm's Law	V = IR. Use to find any one of V, I, R if the other two are known. Works for any resistive element.
Three power formulas	$P = VI = I^2R = V^2/R$. All equivalent. Choose based on which two quantities are known.
Series resistance	R_total = R1 + R2 + R3 … Same current everywhere. Voltage divides proportionally to resistance.
Parallel resistance	1/R_total = 1/R1 + 1/R2 … Same voltage everywhere. Current divides inversely to resistance.
KVL	Sum of all voltage rises and drops around any closed loop = 0. ΣV_source = ΣV_drop.
KCL	Sum of all currents into a node = sum of currents out. Charge cannot accumulate at a node.

Voltage divider	Vout = Vin × R2/(R1+R2). Used for reference voltages, sensor circuits, signal scaling.
MCB type selection	Type B: resistive loads (lighting). Type C: motors, transformers, fluorescent. Type D: high inrush.
Protection sizing	I_device ≥ I_design (carry load) AND I_device ≤ I_cable (protect cable from overheating).

What Is Next — ELEC-010

ELEC-010: Single Phase Systems — Single-phase AC power in depth: RMS voltage and current, instantaneous vs RMS values, AC power factor, apparent power (VA), real power (W), and reactive power (VAR). Single-phase motor connections, single-phase transformer ratios, and single-phase wiring in residential and light commercial buildings.

Resource	Details
ELEC PDF Series	ELEC-001 through ELEC-030 — ayetechub.com/pdfs.html
Engineering Formulas	Free formula reference card — ayetechub.com/downloads/engineering-formulas.html
Community	Telegram: t.me/ayetechub